

Spatial expression heatmap of 12 shared stage-associated PPI hub genes

Expression is standardized within each gene across Normal, ADM, and PDAC regions

Normal

ADM

PDAC

TOP2A

CCNA2

ITGA2

STAT1

ANLN

MKI67

ACTB

CENPF

CASP3

FOXM1

CCNB1

PCNA

Row Z-score

4

2

0

-2

